Azaan Noman

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EDUCATION

Seven Lakes High School | Katy, TX Expected May 2028

Academic interests: Machine learning, autonomous systems, neuroscience, and biomedical engineering.

RESEARCH EXPERIENCE

FlyWire - Cross-Individual Connectome Alignment

Prof. Wayne Hayes, UC Irvine | FlyWire (Princeton Neuroscience Institute) | In progress

- Conduct **topological alignment** of fly-brain networks with **SANA**, alongside **NBLAST** comparisons.
- Investigate which neurons have **one-to-one cross-individual counterparts** and which lack unique correspondence.

sEMG Feature Transferability for Cross-User Hand-Gesture Classification

Co-author | BMES 2026 presentation forthcoming | Research code

- Benchmarked **3 classifier families** using **21 feature types**, **128 channels**, and **2,688 predictors** from **18 participants**.
- Evaluated cross-user generalization through **378 random-forest model fits** and **leave-one-subject-out validation**, ranking features by held-out balanced accuracy.

CALIBER - Context-Aware Neural Network Calibration | *In progress*

- Developing a hypernetwork-inspired calibration layer to adapt trained neural networks to new users with limited contextual data.

Classically Simulable Quantum Circuits for ML Benchmarking

Prof. Stefanos Kourtis (Université de Sherbrooke; Mila - Québec AI Institute) | In progress

- Investigate restricted quantum circuits for ML and optimization, focusing on **classical simulation**, predictive performance, and computational cost.
- Technical reference: Gince et al., Fermionic Machine Learning (2024): **matchgate circuits**, free-fermion simulation, and SVM quantum kernels.

TECHNICAL & PRODUCT EXPERIENCE

Irontrace - Cost-Aware Machine Learning for Hydraulic Inspection (GitHub)

- Developed a **TabPFN 3.5** pump-leakage classifier using **70 features**, **14 sensors**, and **2,205 recorded cycles**.
- Built budget-aware sensor selection with a ridge value model and a FastAPI/React/SQLite console; validated with **52 backend tests** and **8 live API acceptance groups**.

DroneLab - AI Training & Evaluation for Drone Control (GitHub)

- Built a physics-based simulation environment to **train drone-control policies**, combining Rapier dynamics with TensorFlow.js **behavior cloning**.
- Implemented reproducible flight episodes, recording of up to **14,400 transitions per run**, and held-out evaluation against scripted and random controllers.

Axiom - Quantitative Research & Backtesting Platform

- Built an Electron/TypeScript/SQLite research application integrating market data, signal scoring, ML outcome tracking, AI-assisted research, and Discord alerts.
- Developing a Python/XGBoost ranking pipeline with **walk-forward validation**, transaction-cost modeling, and held-out evaluation; research and simulation only.

Worktree - Collaborative AI Developer Workspace (Website)

Co-founder | Product Strategy

- Researched multiplayer AI workflows and translated context fragmentation, overlapping agent work, and visibility concerns into priorities for **shared workspaces, conflict awareness, and local execution**.

CalcQuest - Adaptive Calculus Learning Game | *Junior Developer*

- Developing **machine-learning personalization** for AP Calculus AB/BC practice based on learner progress; also contributing to game design and visuals.

ENGINEERING & COMPETITIONS

RoboSpartans - FIRST Robotics Competition, Team 4639 2026 season

Electrical Engineering | Two-year team member

- Redesigned robot electrical layouts for major structural changes, reorganizing power distribution, CAN wiring, motor controllers, sensors, and cable routing.
- **2026 team results: 3rd of 31** in Dripping Springs qualifications; **District Event Finalist** and **Innovation in Control Award** recipient.

Destination Imagination - Engineering Challenge 2024

*Global Finals | **3rd Place Globally (Team)***

- Created complete blueprints and redesigned a classical trebuchet into a **transforming mechanism**, targeting greater mechanical effectiveness and efficiency.

Pumps & Pipes - ChemE Car | Grand Champion (Team)

- Designed a chemical-reaction-controlled car **from scratch**, integrating mechanical design, chemical formulations, and electrical controls.

WORK EXPERIENCE

Yo Jing Swim Academy | *Swim Instructor* Present

- Teach children competitive swimming **12 hours per week**, providing clear instruction and technique-focused feedback.

LEADERSHIP & SERVICE

Hearts for Elders | *Head of Chapter Management*

- Coordinate volunteers and senior-care initiatives across **25+ global chapters spanning 6 continents**.
- Organization-wide impact: **1,150+ elders reached**, **200+ volunteers**, and **$10,000+ raised**.

Mecclabs | *Mechanical Co-Director*

- Design multidisciplinary engineering camp curriculum; helped generate **nearly $20,000** through educational programs.

Technology Student Association | *Officer*

- Help prepare **50+ competitors**. Chapter results: **2nd among 125 schools** in the State School Award; **14 national qualifiers**.

SLHS Student Council | *Officer*

- Help lead a **600+ member** organization and support **$100,000+ in annual fundraising** for local shelter meals and supplies.

ADDITIONAL PROJECTS

Treehugger: Built a Discord assistant for contextual Q&A, sourced search, and summaries using Groq, Tavily, and a Cloudflare Workers architecture.

Skyes Travel: Built AI booking automation for a family travel business with **$50,000 in revenue**.

RECOIL-1: Designed a bioinspired silicone-membrane pump in AutoCAD; studying reinforcement for consistent refilling.

FPV & racing drones: Built and flight-tested **3 racing drones**, handling component selection, soldering, assembly, and tuning.

Connectome-driven drone autonomy: Investigating fruit-fly neural circuitry for autonomous navigation and control.

Other ventures: Co-founder of **Morro** (morroapp.vercel.app) - adaptive workflow software; **Polyshield** (polyshield.vercel.app) - AI-agent security.

TECHNICAL SKILLS & ACTIVITIES

Programming & tools: Python, TypeScript, MATLAB, React, Electron, Node.js, FastAPI, SQLite, GitHub, Vite, AutoCAD.

ML & simulation: PyTorch, NumPy, TabPFN, TensorFlow.js, Three.js, Rapier, SANA, NBLAST; LOSO validation and behavior cloning.

Competitive climbing: V7 bouldering; speed personal best **7.4 seconds**; train approximately **4 hours per week**.